CURRICULUM VITAE

NAYYER RAZA

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EDUCATION

McGill University, Montreal, Canada

- **Ph.D.** Physics *January 2022 – Present*
  Advisor: Dr. Daryl Haggard

University of British Columbia, Vancouver, Canada

- **M.Sc.** Astronomy *November 2021*
  Advisor: Dr. Jess McIver
- **B.Sc.** Double Major in Physics and Astronomy with Co-op *May 2019*

Highlighted Graduate Coursework:

- PHYS 645 – High Energy Astrophysics
- PHYS 571 – Physical Cosmology
- PHYS 509 – Theory of Measurements
- ASTR 507 – Planetary Science
- ASTR 405 – Astronomical Data Analysis
- PHYS 643 – Observational Astronomy
- PHYS 520 – Teaching Techniques in Physics
- ASTR 508 – Stellar Astronomy
- ASTR 502 – Astronomical Dynamics
- ASTR 404 – Astronomical Instrumentation and Measurements

SCHOLARLY WORK

Articles Published in Peer-Reviewed Journals

- M. L. Chan, J. McIver, A. Mahabal, C. Messick, D. Haggard, **N. Raza** et al. (2024), "GWSkyNet II: a refined machine learning pipeline for real-time classification of public gravitational wave alerts", The Astrophysical Journal, 972, 50 (10pp.)

- **N. Raza**, M. L. Chan, D. Haggard, A. Mahabal, J. McIver, T. Abbott, E. Buffaz, and N. Vieira (2024), "Explaining the GWSkyNet-Multi machine learning classifier predictions for gravitational-wave events", The Astrophysical Journal, 963, 98 (20pp.)

- **N. Raza**, J. McIver, G. Dalya, and P. Raffai (2022), "Prospects for reconstructing the gravitational-wave signals from core-collapse supernovae with Advanced LIGO-Virgo and the BayesWave algorithm", Physical Review D, 106, 063014 (17pp.)

- Z. Yan, **N. Raza**, L. Van Waerbeke, A. J. Mead, I. G. McCarthy, T. Troster, and G. Hinshaw (2020), "An analysis of galaxy cluster mis-centring using cosmological hydrodynamic simulations", Monthly Notices of the Royal Astronomical Society, 493, 1120-1129

- **N. Raza**, L. Van Waerbeke, and A. Zhitnitsky (2018), "Solar corona heating by axion quark nugget dark matter", Physical Review D, 98, 103527 (17pp.)

Highlighted Conference Contributions

- Conference presentation: “Insights into the predictions of a machine learning classifier for gravitational wave events”.

  Presented as an oral talk at the following conferences:
  - Centre de Recherche en Astrophysique du Quebec Annual Meeting, May 2024 (provincial)
  - Astroinformatics 2023 Conference, October 2023 (international)
  - Canadian Astronomical Society Annual General Meeting, June 2023 (national)

- Conference presentation: “Prospects for reconstructing the gravitational wave signal from core-collapse supernovae in Advanced LIGO-Virgo”.

  Presented as an oral talk at the following conferences:
  - 14th Edoardo Amaldi Conference on Gravitational Waves, July 2021 (international)
  - Canadian Association of Physicists Congress, June 2021 (national)
  - American Physical Society April Meeting, April 2021 (international)

  Presented as a poster at the following conference:
  - Canadian Astronomical Society Annual General Meeting, May 2021 (national)

AWARDS AND ACHIEVEMENTS

- Walter C. Sumner Memorial Fellowship ($6650/yr) *2023-2024*
  Competitive national award to support doctoral studies in Physics at a Canadian university
- McGill Graduate Excellence Award ($1350/yr) *2023-2024*
  Internal award top-up to Sumner Fellowship in recognition of research excellence
- Canadian Space Agency Travel Award ($1850) *2023*
  Competitive national award to fund student participation in space conferences
- Trottier Space Institute at McGill Graduate Fellowship ($8000/yr) *2022-2024*
  Internal award to support graduate studies at the institute
- NSERC Undergraduate Student Research Award ($4500, held at University of Toronto) *2018*
  Competitive national award to partially fund undergraduate research work
- NSERC Undergraduate Student Research Award ($4500, held at UBC) *2017*
- UBC Chancellor’s Scholar Award *2013*
  For an incoming academic average of 95% or more into university

TEACHING EXPERIENCE

Graduate Teaching Assistant, Department of Physics, McGill

- 6 hours/week conducting labs, marking lab reports, preparing weekly quizzes, holding office hours, conducting tutorials, and marking final projects, for the undergraduate courses:

  PHYS 131 – Mechanics and Waves *Fall 2024*
  PHYS 339 – Measurements Lab in Physics *Winter 2023, Winter 2024*
  PHYS 320 – Introductory Astrophysics *Fall 2022*

PHYS 183 – The Milky Way Inside and Out *Winter 2022*

Graduate Teaching Assistant, Department of Physics and Astronomy, UBC

- 10 hours/week independently conducting labs in-person managing ~30 students, including lab preparation and marking, for the following undergraduate courses:

ASTR 311 – Exploring the Universe: Stars and Galaxies *Winter 2021*
ASTR 101 – Introduction to the Solar System *Fall 2020, Fall 2019*
ASTR 102 – Introduction to Stars and Galaxies *Winter 2020*

RESEARCH EXPERIENCE

Department of Physics, McGill *January 2022 – Present*

- Working as a graduate researcher with Professor Daryl Haggard studying and developing machine learning models that can rapidly predict gravitational wave event source classifications.

Department of Physics and Astronomy, UBC *September 2019 – November 2021*

- Worked as a graduate researcher with Professor Jess McIver as a member of the LIGO Scientific Collaboration, studying gravitational waves from core collapse supernovae.
- Optimized the BayesWave analysis algorithm to reconstruct and study the expected gravitational wave features from CCSN as they would be detected in Advanced LIGO (masters thesis).

Department of Astronomy and Astrophysics, University of Toronto *September – December 2018*

- Worked full-time with Professor Marten van Kerkwijk on a project analyzing giant pulses emitted by the pulsar PSR B1937+21 using new data obtained at the Aricebo observatory.
- Identified these giant pulses from the raw baseband data and verified that they encode the impulse response function of the interstellar medium. Used the giant pulse population as a whole to infer lensing properties of the scattering screen.

Department of Physics and Astronomy, UBC *January – August 2018*

- Worked full-time on a joint project with Professor Ariel Zhitnitsky and Professor Ludovic Van Waerbeke, exploring the applications of the Axion Quark Nugget (AQN) dark matter model to phenomena in the solar atmosphere.
- Worked out the relevant solar system dynamics involved in the nugget-corona interaction, programmed and performed the numerical simulations, interpreted the results and co-authored the published paper.

Department of Physics and Astronomy, UBC *May – August 2017*

- Worked full-time with Professor Ludovic Van Waerbeke on a project analyzing galaxy cluster mis-centering and its effects on cluster mass profiles using the BAHAMAS hydrodynamic simulations.
- Quantified both the physical offset distributions between different observational centroids and the biases in mass reconstruction by fitting known empirical density profiles, and co-authored the published paper.

OUTREACH AND COMMUNITY WORK

McGill Graduate Association of Physics Students (MGAPS) *2023 – Present*

- Representative/delegate for physics graduate students in the McGill Teaching Assistants (TA) union. Successfully negotiated for higher wages, increased protections, and fairer working conditions for McGill graduate TAs as part of the negotiations team for a new 4-year contract.
- Core member of the team that surveyed the financial situation and mental health of the ~200 graduate students in the physics department in 2023, producing an internal report which was used to successfully advocate for a significant increase in the physics graduate student stipend.

Astronomy Outreach

- Appeared as a guest on a YouTube science channel called "*Kainaat* Astronomy in Urdu", run by an astrophysics professor to promote astronomy education in South Asia, and presented my research in my native language of Urdu. *2023*
- Helped run an activity related to life on exoplanets for elementary school age kids at the Montreal Rio Tinto Alcan Planetarium as part of AstroFest. *2022*
- Appeared as a guest on a CBC national science radio show called "Quirks and Quarks" and answered listener questions about exploding stars. *2020*
- Helped run activities for the public gathered at UBC to observe the near-total solar eclipse. *2017*

UBC Physics Olympics *2016*

- Assisted the head organizer of the Pulleys Lab event in setting up and running the event multiple times throughout the day, for the 60 high school teams participating.

UBC Reading Week Projects *2015, 2016*

- Macdonald Elementary School: Mentored a sixth grade student to create a science fair project on the theme of Space, helping them develop an engaging project and work on presentation skills.
- Mount Pleasant Elementary School: Acted as a leader and mentor to a group of fifth graders, and helped them build a complex machine using simple machines such as levers, pulleys and axles.